AI Supported Design Process Summary

Matcha Machine Design Process

Keywords:

- Daily Ritual
- Clean Energy
- Preparation Anxiety
- Measurement Accuracy
- Self-Cleaning
- Minimal Effort

Selected Impactful Prompts:

AI Role : Clarifier

Prompt 1 — Context Clarification
"Ask me 10 precise questions to define the real usage context of a hot beverage device including space constraints, safety habits and sensory expectations."

Prompt 2 — User Profile Evaluation
"Would defining both students and young professionals as target users create conflicts in later design decisions?"

Prompt 3 — Evidence Card Generation
"Based on my interview answers, create evidence cards summarizing user behaviors, insights and quotes."

Prompt 4 — Insight Prioritization
"Create an Impact × Frequency matrix to prioritize insights from my interviews."

Initial Design Direction: A seamless, single-action matcha maker that automates the ritual, delivering a zero-hassle experience with minimal user effort.

Phase 1 — Context Exploration
Hot beverage preparation behaviors were investigated.
AI was used as a **Clarifier** to generate contextual questions about user behavior, safety, and environment.
These questions helped structure the initial research and identify missing knowledge areas.

Phase 2 — Target User Analysis
Initial target group: users who treat kitchen activities as a ritual.
Research revealed that this group is already well served by existing products.
The design direction shifted toward **matcha consumption**, where a market gap was identified.

Phase 3 — Market & Cultural Research
Matcha history, rituals, and global trends were analyzed.
Findings showed a conflict between **traditional preparation rituals and modern convenience needs**.

Phase 4 — User Interviews
Interviews with young users revealed key pain points:

- Cleaning effort
- Measurement anxiety
- Whisking effort
- Time pressure

These insights became the basis for the design opportunity.

Phase 5 — Insight Structuring with AI
AI transformed interview data into **Evidence Cards**.
This allowed complex data to be broken into actionable design insights.

Phase 6 — Insight Prioritization
An **Impact × Frequency Matrix** was used to identify the most critical problems:

- Post-consumption cleaning burden
- Preparation effort
- Dose anxiety
- Biological timing constraints.

Phase 7 — Problem Framing & Reframing
AI was used as a **Re-framer** to reinterpret the problem from multiple perspectives:

- Efficiency
- Cognitive load
- Identity & lifestyle expression.

AI Response Summaries:

Clarifier responses

- Generated contextual research questions
- Helped identify missing research areas.

Critique responses

- Highlighted conflicts between student and professional personas.

Expander responses

- Converted interviews into structured evidence cards.

My Reflections:

- AI helped organize scattered ideas into structured insights.
- Interview data became easier to analyze when translated into evidence cards.
- Using prioritization matrices prevented personal bias in design decisions.

Accepted:

- Evidence cards created by AI
- Impact × Frequency prioritization
- Focus on preparation friction.

Rejected:

- Initial ritual-focused niche kitchen user group
- Dual persona strategy (student + professional).

Transformed:

- Traditional matcha ritual → fast everyday routine
- Manual whisking → automated preparation system

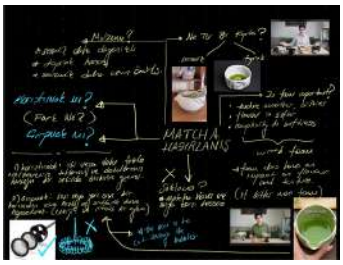
INTERNAL (Product & User)	STRENGTHS • Clean, long-lasting energy without the crash (L-theanine) • The aesthetic, "Instagrammable" at-home cafe experience • Strong alignment with rising health and wellness trends	WEAKNESSES • Preparation hassle (using a chasen, washing extra bowls) • High price barrier and import costs • Earthy taste profile is unfamiliar to the local palate
EXTERNAL (Market & Environment)	OPPORTUNITIES • Huge market gap for practical, all-in-one preparation tools • Direct-to-Consumer subscription models • Promoting alternative daily uses (smoothies, oatmeal)	THREATS • Deeply rooted traditional black tea and coffee culture • Low-quality matcha in the market ruining first impressions • Inflationary pressures leading to cuts in premium spending

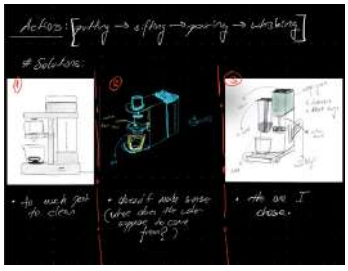
COMMON ASPECTS	DIFFERENT ASPECTS
1) Healthy Alternatives and the Perception of "Clean Energy"	**1) Functionality vs. Emotional Ritual**
All respondents viewed matcha as a much healthier alternative to coffee or energy drinks. Instead of the addictive effects of coffee or the artificial flavors/palpitations of energy drinks, matcha's antioxidant properties, metabolism-boosting effects, and natural ingredients were a common motivation.	Matcha's place in daily life varies greatly from person to person. For some, it's simply a caffeine/energy booster used "when needed," while for others, its aroma and preparation create a special moment of "pause" and enjoyment during the day.
2) A Shared Dislike of the Cleaning and Preparation Process	**2) Identity and Image Reflection**
The biggest pain point for users is definitely the hassle of the process. The lengthy whisking process (1st person), the hassle of washing dishes (2nd and 4th people), or the need to run the machine empty/clean it (3rd person) detracts from the experience. Everyone wants to achieve results in seconds and not have to deal with any cleaning burden afterward.	For some users, matcha consumption is a lifestyle statement. One person sees it as part of their identity, projecting a "healthy girl" image, another as a "clean boy" image. However, for the second and third individuals, what they drink has nothing to do with the image they project.
3) Arome and Flavoring	**3) Preparation Method and Specific Barriers Encountered**
For this group, the pure matcha taste isn't always sufficient or appealing. They tend to consume it with flavors like strawberry (2nd person) or vanilla (1st person). Also, while they appreciate the convenience of capsules, they don't want to give up the intense, natural aromatic experience offered by powdered matcha (3rd and 4th people).	Everyone's system and therefore their problem is different. The first person complains about not being able to get the right consistency/ratio with a metal whisk. The third person is stuck on the limitations of the capsule machine (poor quality matcha powder). The fourth person gets stressed out adjusting the powder/water ratio while calculating cost and caffeine.

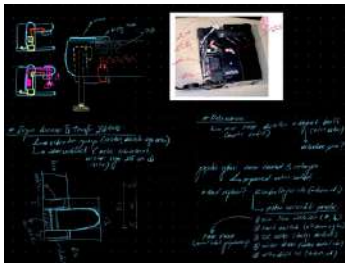
Insight	Impact	Frequency	Selection
Tüketim sonrası temizlik yükü ve bulaşık stresi	5	5	Seçildi (Top 1)
Hazırlık aşamasındaki bekleme ve çırpma eforu	4	5	Seçildi (Top 2)
Hız için geleneksel lezzetten/kaliteden ödün verme hissi	3	4	Elendi
Pahalı tozu ziyan etme ve doğru kıvamı/oranı tutturamama kaygısı	5	4	Seçildi (Top 1)
Biyolojik zamanlama kısıtlaması (Günün geç saatlerinde kafeini atamama)	4	4	Seçildi (Top 2)
Kirli ekipmanların yarattığı koku ve estetik/duyusal rahatsızlık	4	3	Elendi

PRIORITY FEATURES	
1) Effortless Cleaning (Self-Cleaning / Easy Discharge):	This is the most striking aspect of the design. Having the parts in a closed system that doesn't retain odors, cleans itself instantly when rinsed with water, or is self-cleaning, will completely eliminate users' fear of "having to deal with it afterwards.
2) Automatic Portioning and Consistency Control:	A mechanism is needed to relieve the user of the stress of wondering "how much powder should I add?" and "is the consistency right?". The focus should be on an integrated whisking/mixing technology that takes the ideal amount of matcha powder from the container and blends it perfectly (at capsule speed but with the traditional powder flavor).
3) Aroma Integration:	It's clear the target audience is gravitating towards different taste profiles (vanilla, strawberry, etc.). The design could incorporate a secondary micro-container where natural flavorings or sweeteners could be added alongside the fresh matcha powder.
4) Speed-focused, compact design:	Especially during the morning rush or when energy levels drop in the office (before 2 pm), a holistic form factor that completes the process in seconds and doesn't disintegrate and scatter around is a lifesaver for these users.

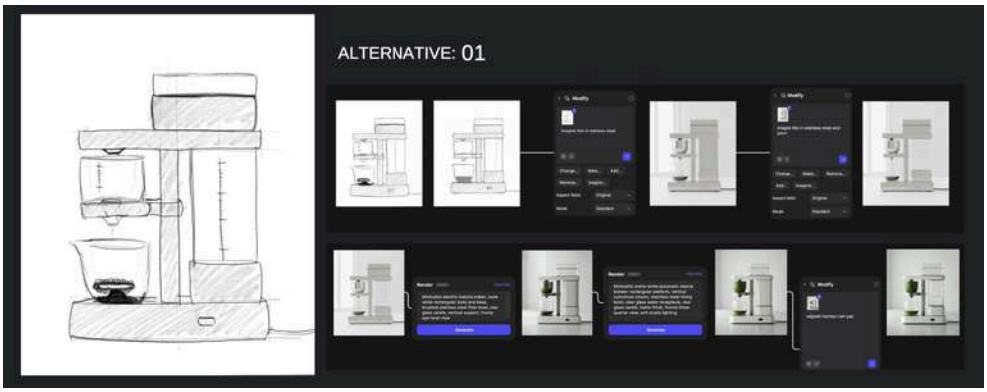
AI Response Summaries:

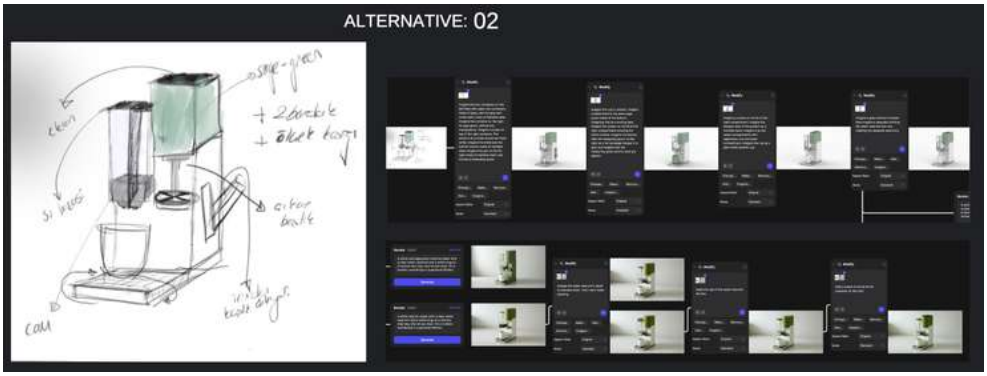


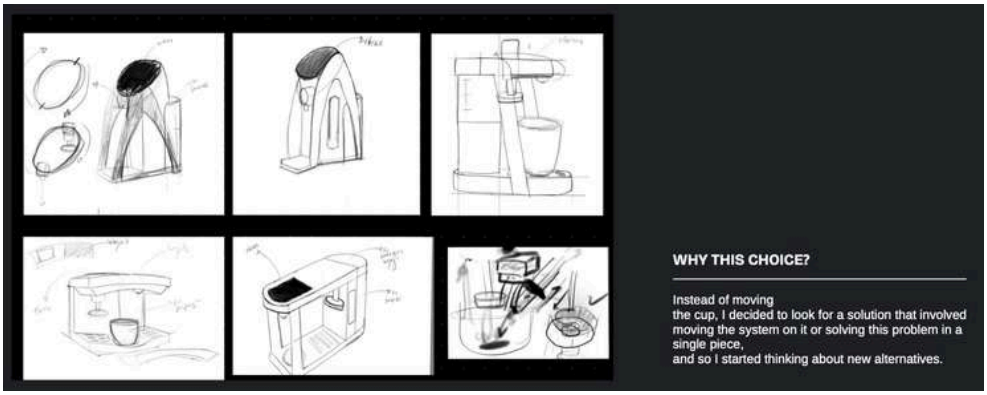


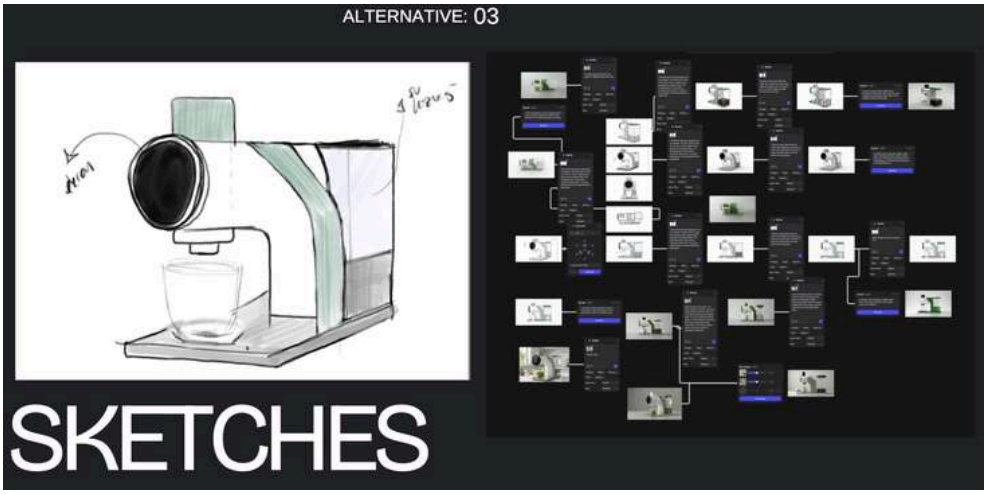


Sketches:

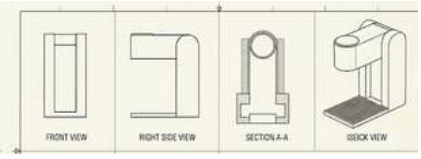


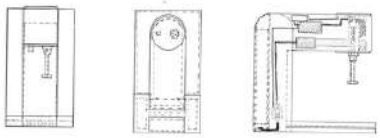


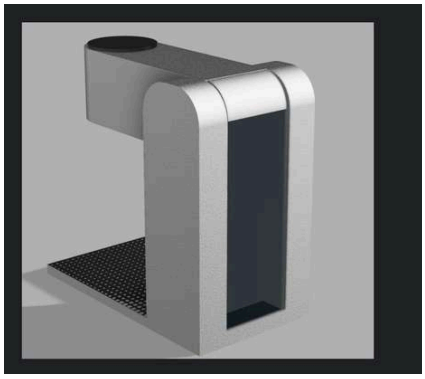


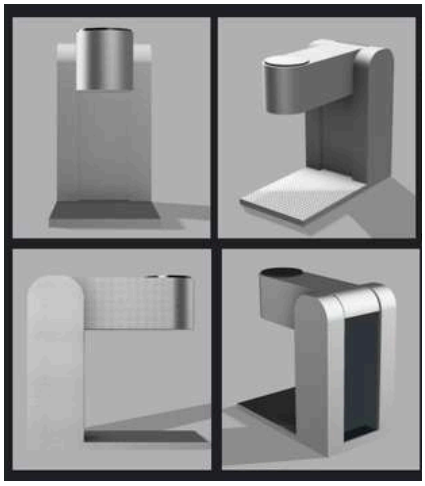


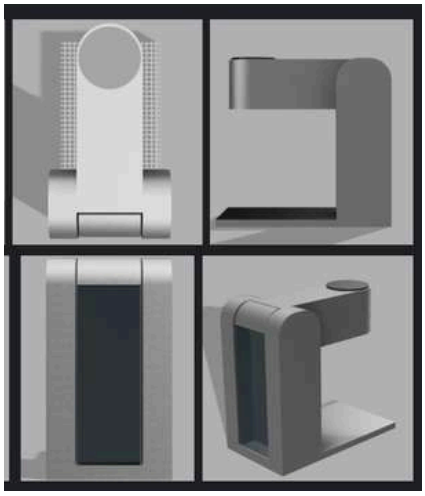
3D Model of The Final Sketch:



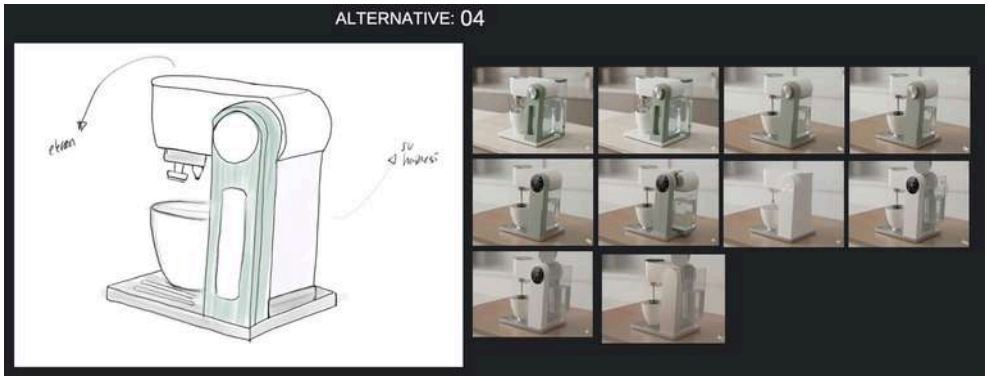








Final Desicion:



- I adapted the hand-squeezed flour sifter mechanism into a spoon-scale measuring tool. Instead of shaking, the user activates the spoon by squeezing a trigger integrated into the handle. combining **measuring**, **sifting**, and **pouring** into one compact action.

Final Design:

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